

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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STG332

Deploying modern applications using Amazon EBS

Trang Luong

(she/her)
Sr Product Manager
AWS

Kevin Liu

(he/him)
Sr Product Manager
AWS



Agenda

- 01 Modernization Journey
- 02 Modernize by Moving to the Cloud
- 03 Modernize with Managed Services
- 04 Modernize with Containers and Serverless
- 05 White boarding session: Containerization with EBS
- 06 Q&A

Modernize your applications to:



Save cost by reducing operations burden and under-utilization of compute and storage

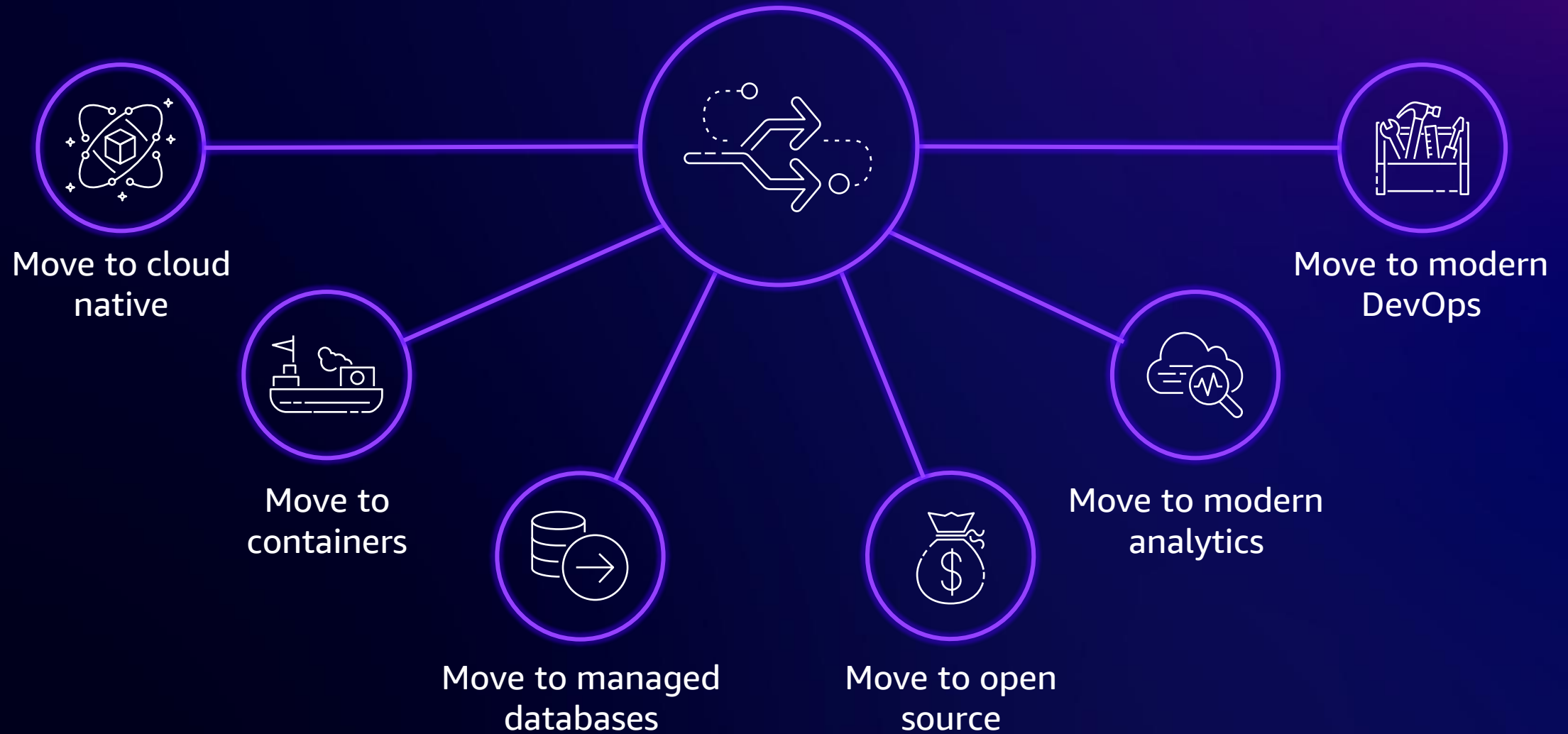


Increase agility by instantly scaling up according to demand



Develop and deploy applications with **greater efficiency**

The modernization pathways



The modernization pathways: Moving to the cloud

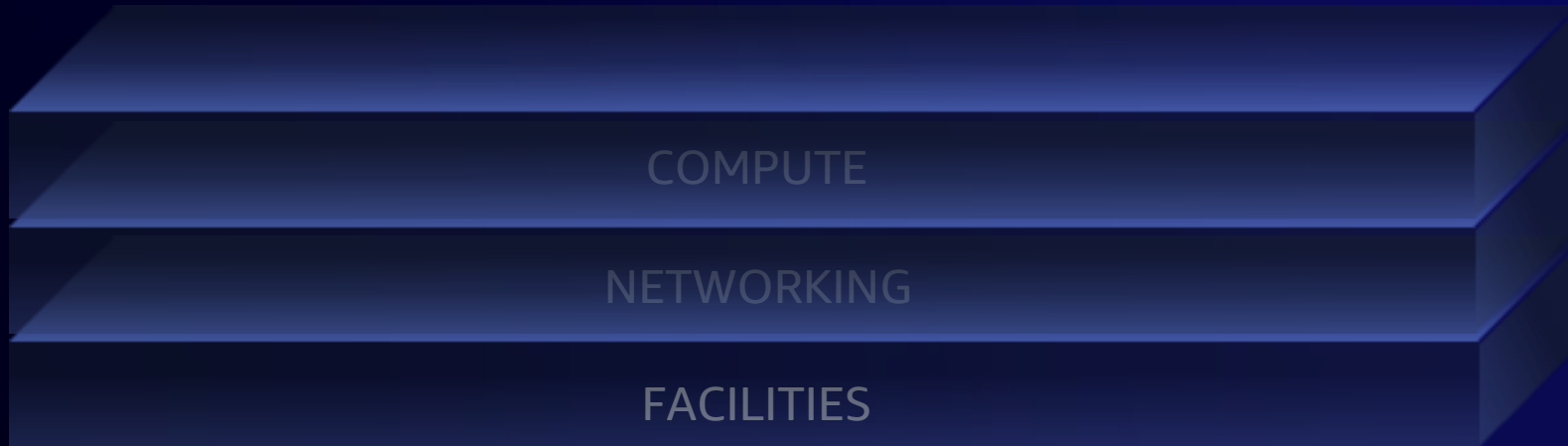


Modernizing infrastructure by moving to the cloud

Managed by
you



Managed by
AWS



Move to the cloud using Amazon EBS

STRAIGHTFORWARD, HIGH-PERFORMANCE BLOCK STORAGE AT VIRTUALLY ANY SCALE



Performance for any workload

- Up to 256,000 IOPS/volume
- 400,000 IOPS/instance
- As low as sub-millisecond latency



Ease of use

Add and remove capacity, or change volume types with Amazon EBS Elastic Volumes



High reliability

- High availability
- Up to 99.999% durability
- Crash consistent snapshots



Virtually unlimited scale

Use a single gigabyte or less, or scale up to petabytes of data



Secure

Encrypt new volumes and data for a region by default with a single setting

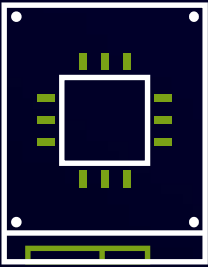


Cost effective

Pay as little as \$0.015 per GB-month for highly cost-effective block storage

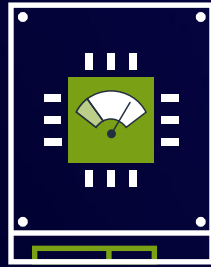
Amazon EBS volume types

SSD



General-purpose
gp3

Balanced price performance for transactional mid-size workloads and boot volumes



Provisioned IOPS
io2 Block Express

Mission-critical, large performance-sensitive workloads requiring sub-millisecond latencies

HDD



Throughput-optimized
st1

Low cost HDD for frequently accessed, throughput-intensive workloads



Cold
sc1

Lowest cost HDD for large volumes of infrequently accessed workloads

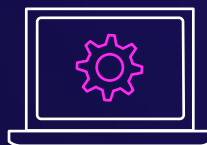
Leveraging Amazon EBS for modern applications

Back up volumes



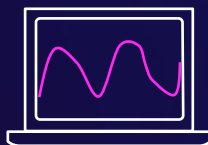
Amazon EBS Snapshots

Automate backups



Amazon Data Lifecycle Manager
AWS Backup

Respond to demand



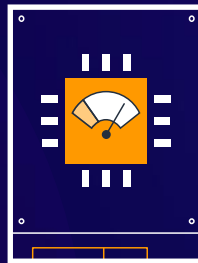
Amazon EBS Elastic Volumes

EBS optimizations in AWS Compute Optimizer

ELIMINATES MANUAL OPTIMIZATION, AVAILABLE AT NO ADDITIONAL CHARGE

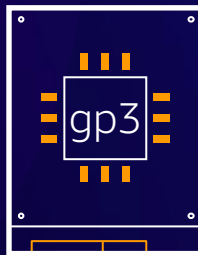


AWS Compute Optimizer



Io2 Block Express

Optimize provisioned IOPS settings



gp3

Right-size gp3 volumes by provisioning IOPS and throughput separately

IOPS and throughput recommendations for gp3 volumes

The modernization pathways: Managed services



Fully managed services on AWS

SPEND TIME INNOVATING & BUILDING NEW APPS, NOT MANAGING INFRASTRUCTURE

Self managed

You

Applications

- Schema design
- Query construction
- Query optimization

Infrastructure

- Automatic fail-over
- Backup & recovery
- Isolation & security
- Industry compliance
- Push-button scaling
- Automated patching
- Advanced monitoring
- Routine maintenance
- Built-in best practices

Fully managed

You

Applications

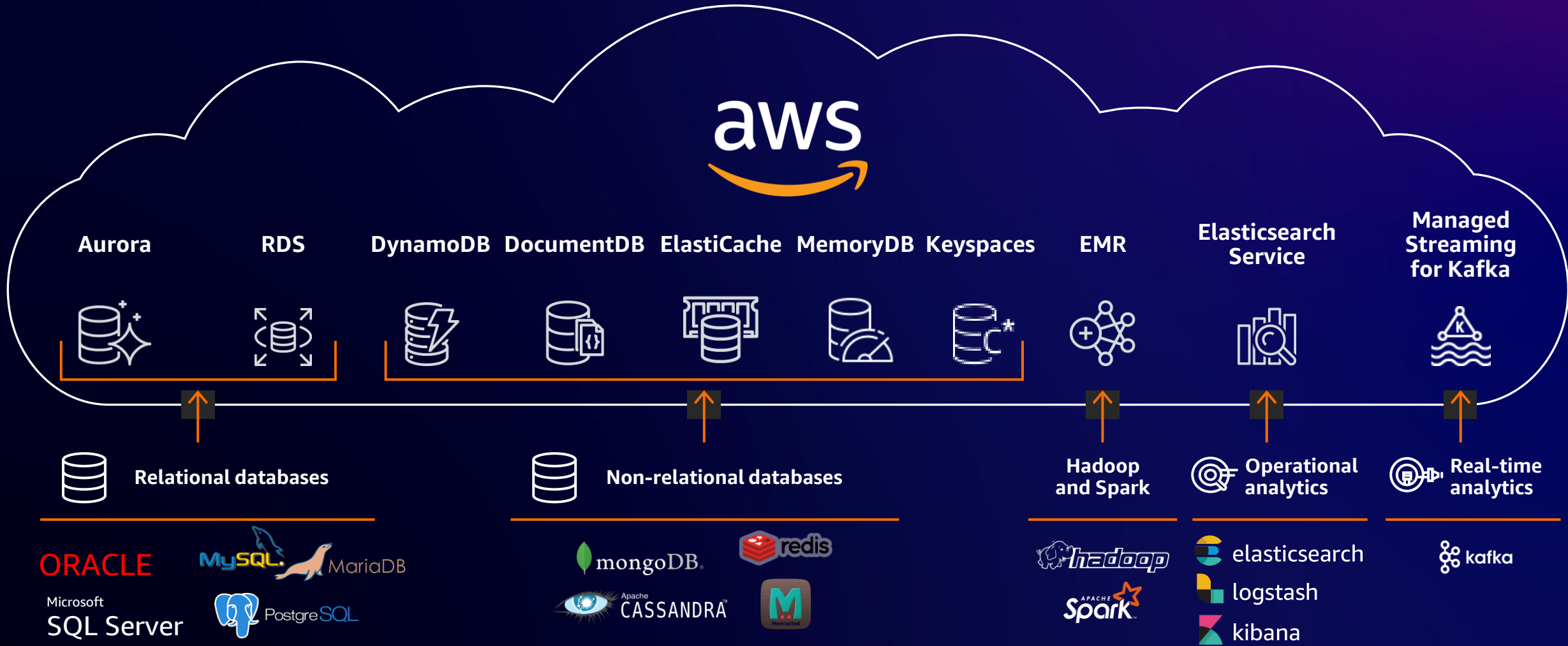
- Schema design
- Query construction
- Query optimization

AWS

Infrastructure

- Automatic fail-over
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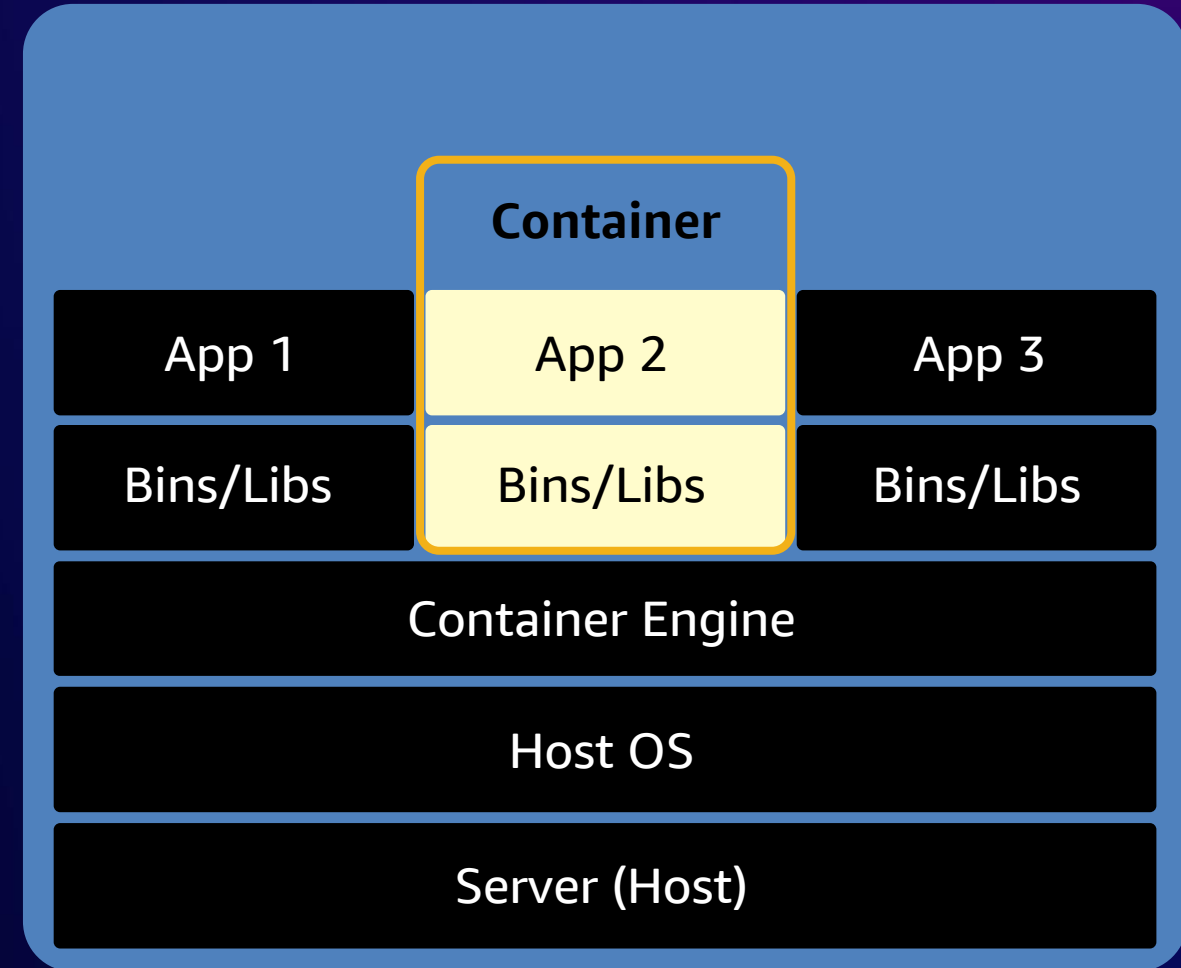
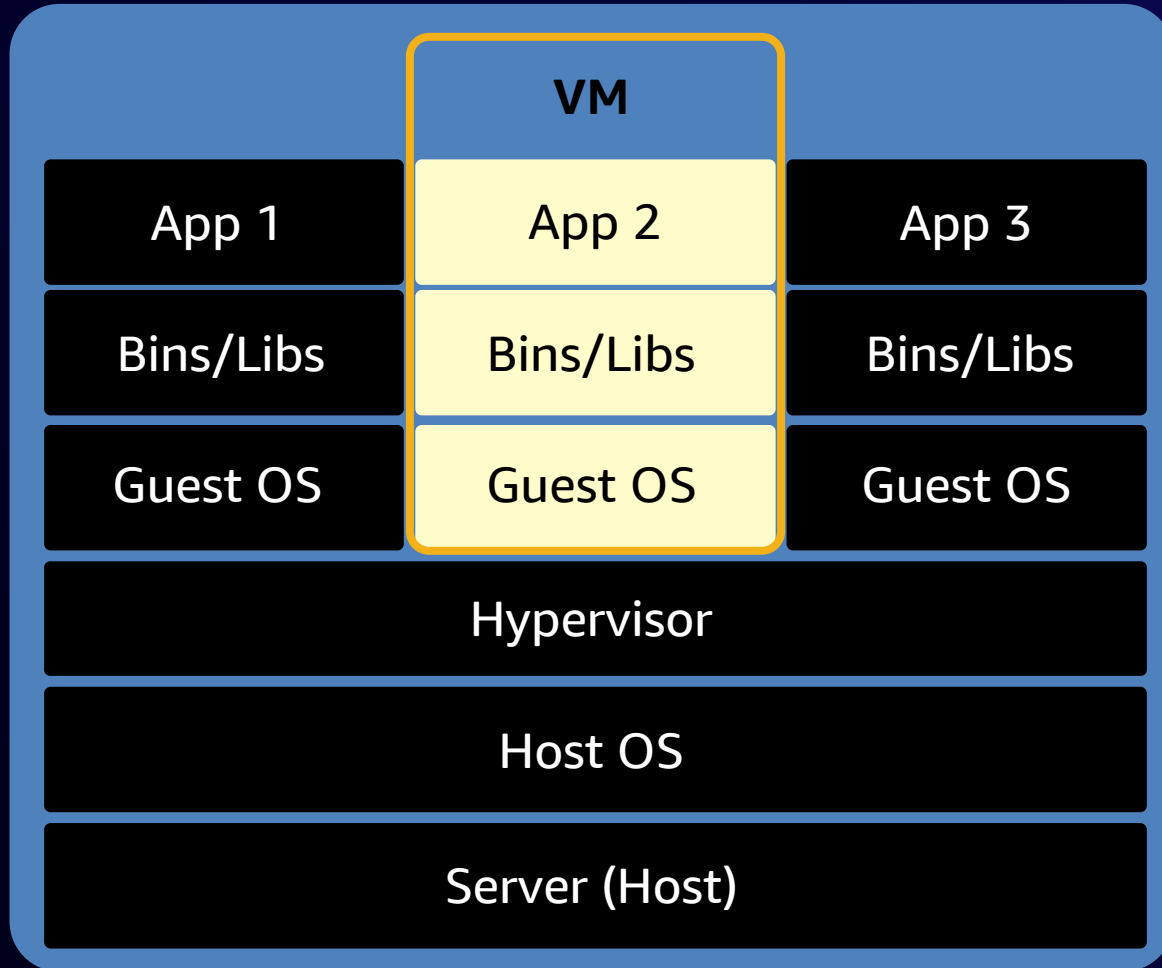
Fully managed database & analytics services



The modernization pathways: Move to containers



Virtual machine versus Container



Benefits of moving to containers

Reduced risk



Uniform security across environment, maintained with automation

Operational efficiency



Reduced operational burden by removing undifferentiated heavy lifting

Speed



Consistent environment improves developer velocity

Agility



Automation increases speed and ease of testing and iterating

AWS container services landscape

Management
Deployment, Scheduling,
Scaling & Management of
containerized applications

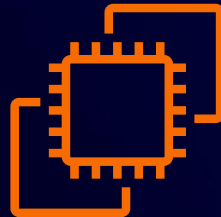


Amazon ECS
**Amazon Elastic
Container Service**



Amazon EKS
**Amazon Elastic
Kubernetes Service**

Hosting
Where the containers run

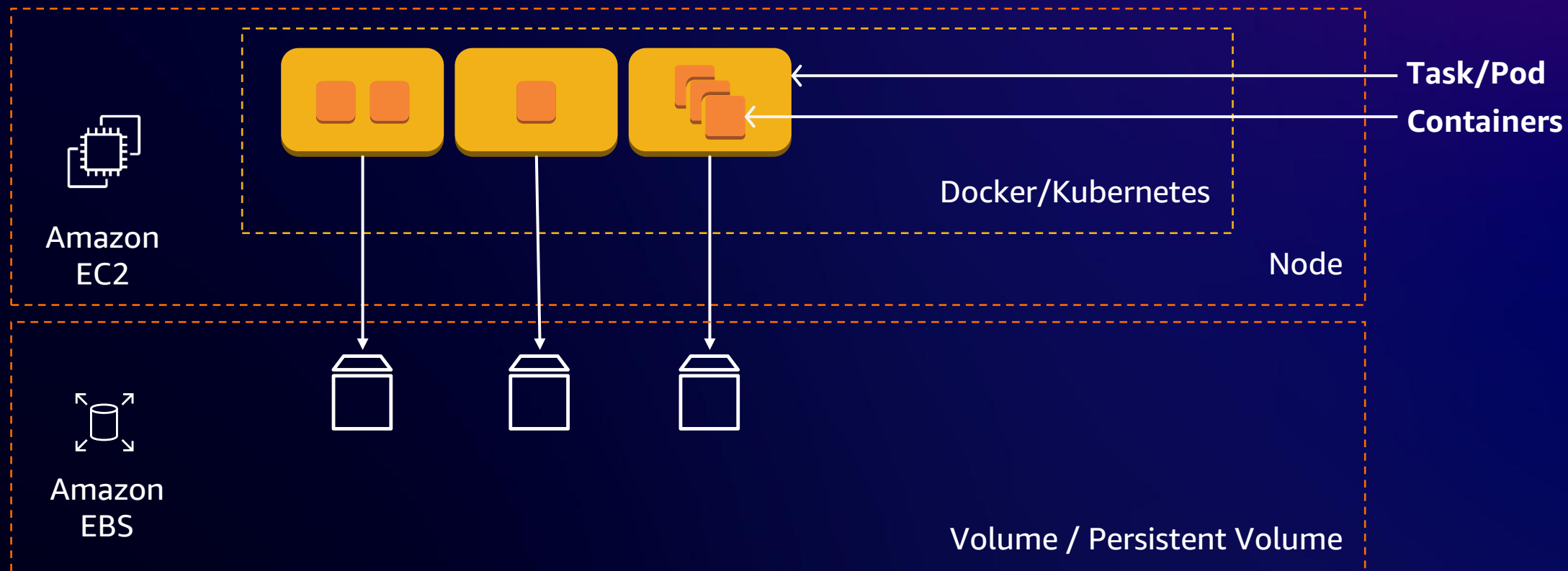


Amazon EC2



AWS Fargate

Using EBS storage for your containers



Newly launched EBS features

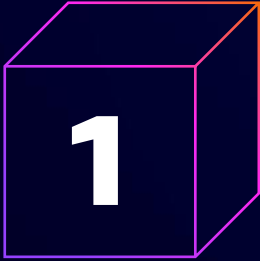


Increased number of
EBS attachments on
new EC2 instances



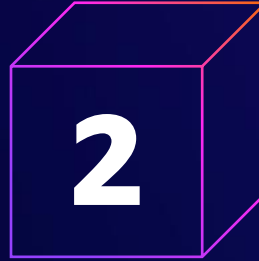
Support full range of
Elastic Volume capability
using EBS CSI driver

Key takeaways



Modernization

Evaluate where your organization is in your modernization journey to adopt the right strategy



Containers

You can adopt containers to continue improving on cost, efficiency, and speed



Benefits of EBS

Leverage new and existing EBS features to save costs and improve deployment agility

Keep learning more about modernization and EBS



Modernize your
applications, drive
growth and
reduce TCO



Blog: Simplifying
Amazon EBS volume
migration and
modification on
Kubernetes using the
EBS CSI Driver



Want to learn more
about EBS?
Other re:Invent
sessions on
Amazon EBS

Thank you!

Trang Luong

luotrang@amazon.com

Kevin Liu

ksl@amazon.com



Please complete the session
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